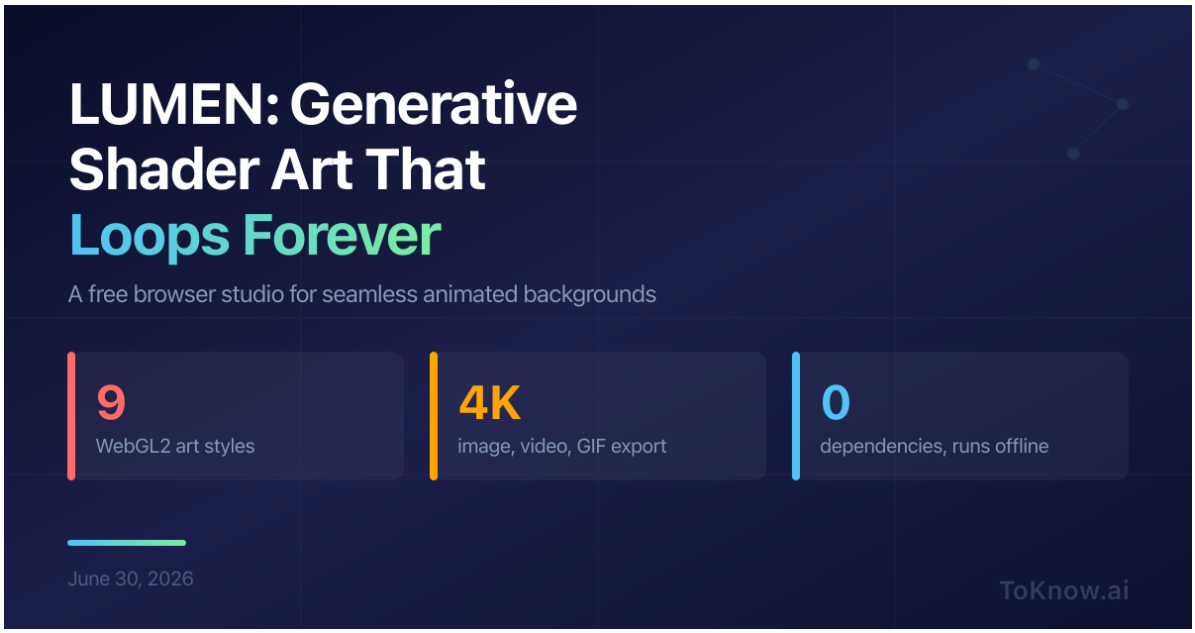
LUMEN: A Free Browser Studio for Perfect-Looping Animated Backgrounds

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Leon Lin built LUMEN, a free generative shader studio that runs entirely in a browser tab, with no neural network involved: the visuals come from math, drawn in real time with WebGL2, the browser's hardware-accelerated graphics engine. There are 9 built-in styles, including liquid chrome, silk, halftone, data glyphs, and reeded glass, plus a synthesizer that mixes 12 "genes" to invent looks beyond that base set. Every animation is a mathematically perfect

loop, sampled around a closed circle so the last frame flows exactly into the first. You can export a still up to 4K, a WebM video, or a GIF, all rendered in the page.

You want a slick animated background for a site header, a video intro, or an album cover, and your options are paying for stock loops or wrestling with heavy desktop software. LUMEN makes one free, with no sign-up. Designs are deterministic, so the same seed always rebuilds the same image, and a share code starting with LMN1 recreates any design exactly. Its gradient-set button spits out 4 to 12 matching variations as a ZIP, enough for a whole website in one click.

What used to need desktop motion software and render time now lives in one static HTML file, and unlike AI image generators such as Ideogram 4.0, it draws seamless motion from math rather than one-shot stills from a prompt. Browser-native tools, WebGL2 for drawing and WebCodecs for encoding video offline, do the whole job locally, with zero dependencies under an MIT license.

Sources:

- LUMEN on GitHub
- LUMEN live studio
- LUMEN documentation
- LUMEN MIT license
- MDN: WebCodecs VideoEncoder

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